LungLogic UI

From breathing requirements to testable control

Capstone update for 4DMedical / Medical Group

Focus: what the interface does, what the simulation shows, and which system requirements still need agreement

22 September 2026

Objective first: make breathing performance visible and testable

OBJECTIVE

Let a user configure a breathing target, see the pneumatic response, and judge whether repeated breaths meet the target before hardware deployment.

01

250 mL per branch

500 mL total system target

02

12–18 bpm

15 bpm nominal; I:E = 1:2

03

30 repeatable cycles

tidal volume within ±5%

04

Independent branches

safe ATM / VAC switching

Prototype scope: one 250 mL branch; dual-branch performance is not yet demonstrated.

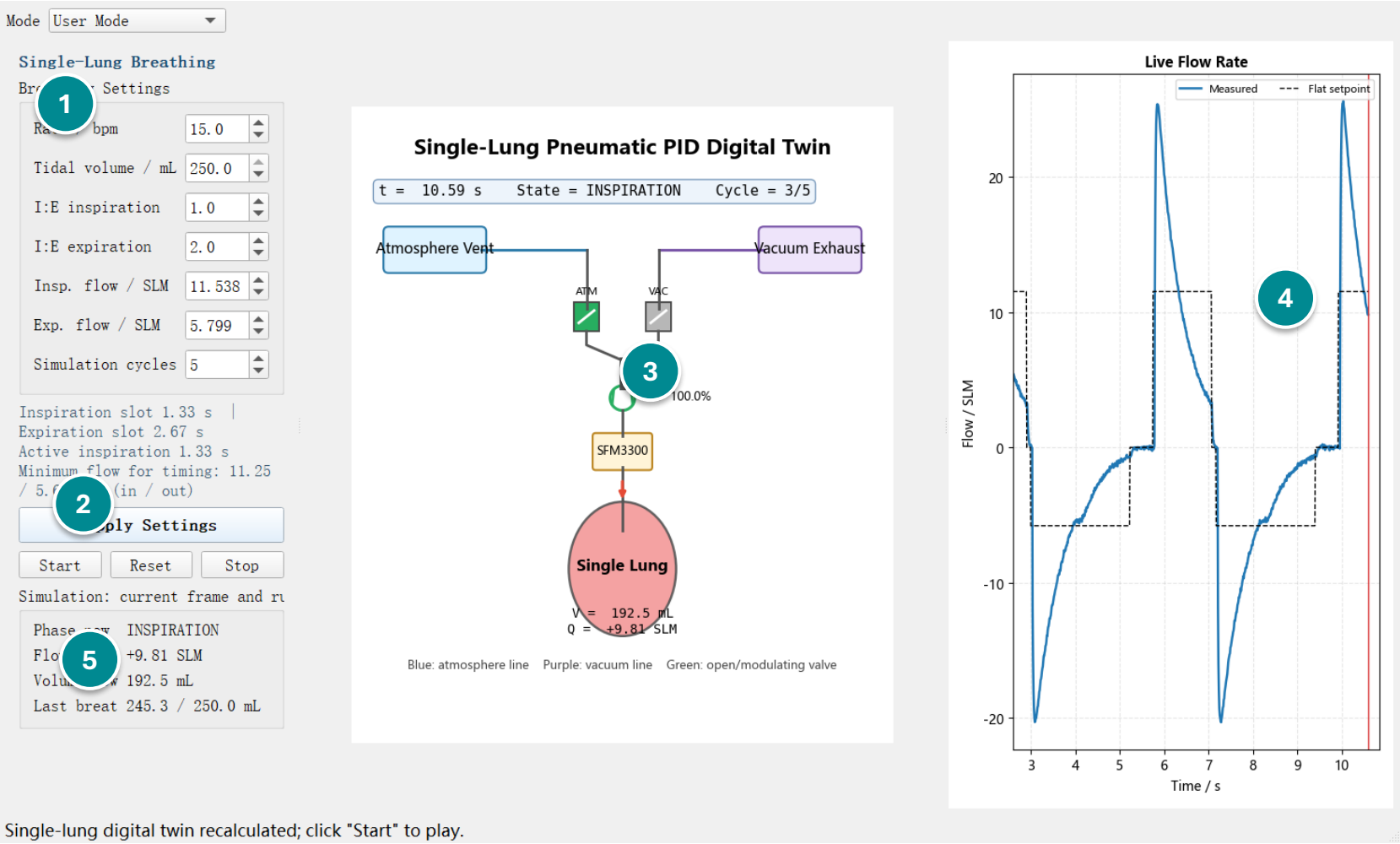
Requirements source: project brief and PROTOTYPE_REQUIREMENTS.md

02

UI walkthrough

One simple mode for setting and reviewing a breath; one technical mode for understanding the controller and assumptions.

User Mode: set a breath, play it back, assess the result



1 **Breathing inputs**

Rate, tidal volume, I:E, flow and cycle count.

2 **Apply & playback**

Recalculate, then start, pause, reset or stop.

3 **Pneumatic state**

ATM/VAC route, PPV opening, phase, flow and volume.

4 **Live flow**

Measured model output against the flat setpoint.

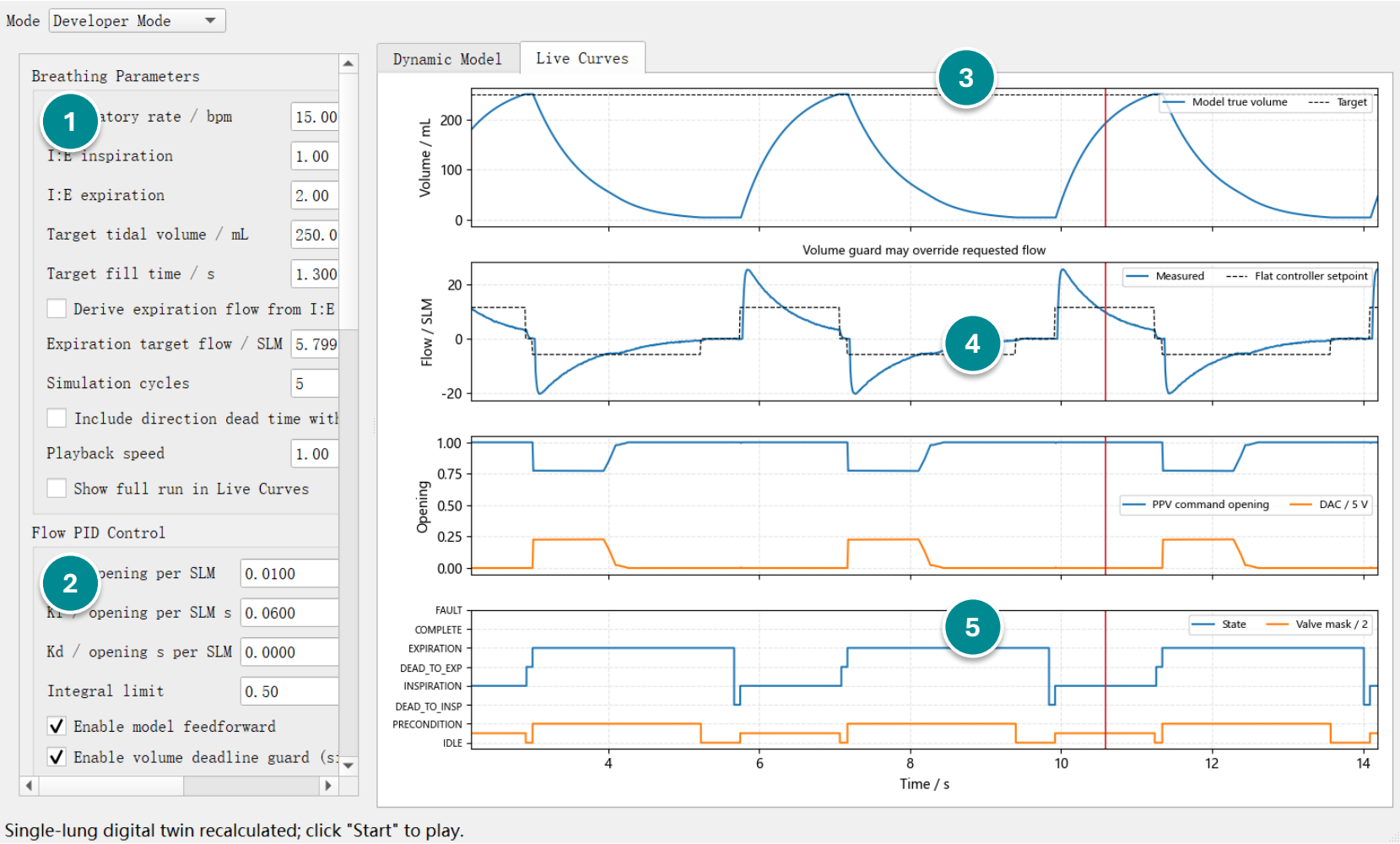
5 **Outcome**

Current frame plus last-breath volume versus target.

The interface exposes clinical-style settings while keeping model assumptions out of the main workflow.

Figure: current single_lung_pid_gui.py User Mode; displayed values are simulated

Developer Mode: expose assumptions and diagnose why a case passes



1 Test definition

Breathing timing, target volume and run length.

2 Controller options

PID, feedforward, Smith prediction and volume guard.

3 Volume evidence

True model volume and target over repeated breaths.

4 Flow evidence

Sensor-model flow reveals overshoot and missed setpoints.

5 Actuation & state

PPV opening, DAC command, route and phase timing.

A passing volume is interpreted together with flow tracking, saturation, timing and valve state.

Figure: current single_lung_pid_gui.py Developer Mode; displayed values are simulated

03

Current status

What is implemented, what the 30-cycle study shows, and why the nominal 15 bpm case still needs a requirements decision.

The UI is implemented; the physical closed loop is not yet verified

AVAILABLE NOW

- ✓ User and Developer workflows
- ✓ Input checks and timing warnings
- ✓ Flow, volume, command and state plots
- ✓ **Repeatable 30-cycle simulation studies**
- ✓ Saved configuration and playback

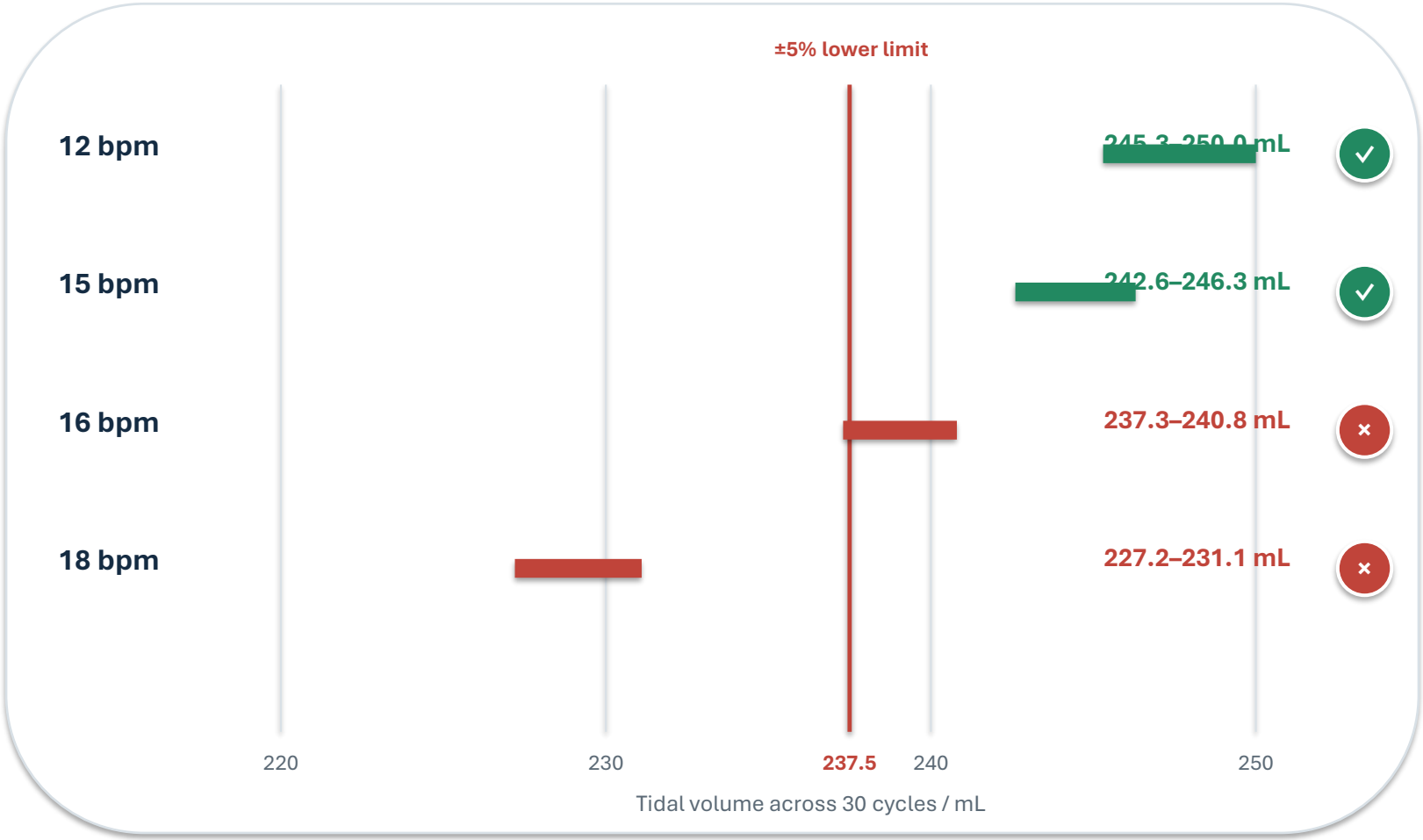
NOT YET DEMONSTRATED

- – Physical PPV actuation from this controller
- – SFM3300 feedback used by the PID
- – Synchronized pressure acquisition
- – 30-cycle hardware repeatability
- – Independent two-branch operation

Use the UI today for simulation, requirement discussion and experiment planning—not hardware acceptance.

Evidence: source code, STM32 demo scope, PROTOTYPE_REQUIREMENTS.md

At 15 bpm, thirty simulated breaths pass with almost no margin



15 bpm

242.6–246.3 mL

Passes the ±5% simulation band

Critical caveat

PPV is fully open for ~100% of inspiration.

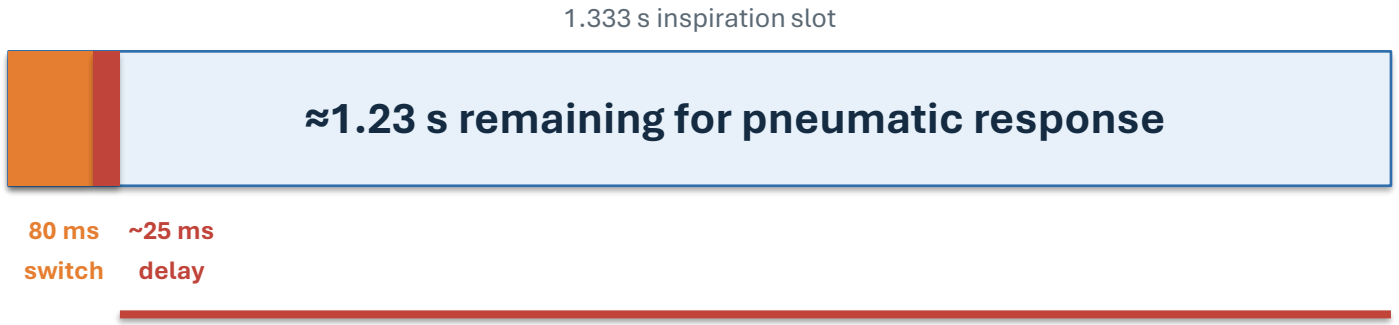
No meaningful throttling margin remains.

This is a model result under stated assumptions, not evidence of installed-path performance.

Data: output/requirement_benchmark/single_lung_30_cycle_benchmark.md; guard on, assumed plant

15 bpm is a requirement trade-off, not just a PID tuning problem

Available inspiration time at 15 bpm, I:E = 1:2



Assumed full-open reach: 1.30 s

Result: the controller is saturated before it can create comfortable margin.

DECISION NEEDED FROM 4DMEDICAL

1. **Must 15 bpm, 250 mL and I:E 1:2 all be fixed simultaneously?**
2. **Is ±5% required for every breath, and is a high early-flow pulse acceptable?**
3. **If the path saturates, should we change timing, target volume or hardware capacity?**

18 bpm under the current assumed plant: 227–231 mL; PID gain changes cannot overcome full-open saturation.

When the PPV is already fully open, software cannot create additional pneumatic flow.

Timing and saturation evidence: benchmark and 18 bpm trade-off studies; installed path remains unmeasured

04

Future

Calibrate the installed pneumatic path, close the hardware loop, and validate against the agreed acceptance test.

Future work: replace assumptions with measurements, then validate

1 Calibrate the path

Full-open ATM/VAC response

Fixed-opening PPV sweep

Synchronize Q(t), V(t), P(t), commands

Measured model

→

2 Close the real loop

Drive PPV and routing valves

Feed SFM3300 data to controller

Verify timing and safety interlocks

Hardware closed loop

→

3 Validate performance

30 cycles at agreed nominal/boundary

Report volume, flow and saturation

Then test independent dual branches

Requirement evidence

Success is a measured 30-cycle result against an agreed requirement—not a favourable simulation trace.

Proposed experiments: experiment_design_speaker_notes.md and PROTOTYPE_REQUIREMENTS.md